Metal Price Forecasting using Machine Learning

FEATURE ENGINEERING & REGRESSION MODELING

PYTHON STACK	DOMAIN	METHOD	OUTCOME
Pandas, NumPy, Scikit-learn, Matplotlib	Commodity trend analysis	Time-series regression workflow	Forecasting intelligence for business decisions

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BUSINESS CONTEXT

Problem Statement

Why metal prices are hard to forecast

Inflation shifts real purchasing power and distorts historical nominal trends.

Industrial demand changes with construction, manufacturing, and energy cycles.

Geopolitical events create sudden disruptions in supply chains and investor sentiment.

Economic uncertainty amplifies volatility and makes rule-based analysis unreliable.

Project objective

Build a machine learning-driven forecasting system that learns from historical time-series behavior and predicts future metal price movement with structured regression models.

ML
Forecasting engine

TS
Time-series signals

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DATA FOUNDATION

Dataset & Challenges

METALS ANALYZED

Gold, Silver, Aluminium, Nickel, Uranium

A multi-metal setup captures different market dynamics, from safe-haven behavior to industrial cyclicality.

TIME HORIZON

1992–2021

Roughly three decades of historical observations enable trend detection, regime changes, and seasonality analysis.

CORE DIFFICULTIES

Missing values, inconsistent timestamps, volatility, inflation-adjusted interpretation

These issues can mislead models unless the data is cleaned and converted into stable predictive signals.

🥇 Precious metals

🏭 Industrial demand

⚡ Energy-linked shocks

📉 Noisy price behavior

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TRANSFORMATION LAYER

Data Preprocessing & Feature Engineering

Preprocessing

- Missing value handling
- Date-time conversion
- Chronological sorting
- Data consistency checks

Feature engineering

- Lag features for short-memory patterns
- Rolling mean features for local trend smoothing
- Percentage returns for relative movement capture
- Year, month, and quarter extraction for calendar effects

PURPOSE

Convert raw historical records into structured predictive signals that machine learning models can interpret consistently.

MODEL SELECTION

Machine Learning Approach

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Linear Regression

Acts as the baseline trend prediction model. It captures broad linear market behavior and gives an interpretable benchmark for long-term directional movement.

🌲

Random Forest Regressor

Handles nonlinear relationships and interaction effects more effectively. It is better suited to volatile financial behavior where simple linear assumptions often break.

TARGET TYPE

Continuous numerical prices

REASON FOR REGRESSION

Forecast exact value levels instead of only up/down classes

MODELING LENS

Baseline + nonlinear ensemble comparison

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PIPELINE VIEW

Model Training Workflow

Historical Data	Feature Engineering	Train-Test Split	Model Training	Prediction & Evaluation

Evaluation metrics

MAE	RMSE	R^2
Average absolute error	Spike-sensitive error	Explained variance

Important consideration

Time-series order was preserved by avoiding random shuffling during training, which keeps the learning process aligned with real forecasting conditions.

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INTERPRETATION

Key Insights & Observations

Linear Regression

- ✔ Strong long-term trend learning
- ✔ Accurate when market behavior is relatively stable
- ⚠ Less effective during abrupt spikes and structural breaks

Random Forest

- ✔ Better nonlinear pattern recognition
- ✔ Improved performance on volatile metals
- ✔ Captured medium-term fluctuations more effectively

CRITICAL INSIGHT

Extreme market shocks remain difficult for traditional regression models to predict accurately, even after strong feature engineering.

VALUE CREATION

Business & Industrial Impact

Manufacturing

Procurement planning, raw material cost forecasting, and reduced pricing surprises.

Investment & Finance

Commodity trend analysis, inflation hedge evaluation, and better portfolio timing support.

Supply Chain & Risk

Pricing strategy optimization, volatility monitoring, and scenario-aware planning decisions.

STRATEGIC VALUE

Supports data-driven decision-making and business planning with forward-looking pricing intelligence.

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EXPANSION PATH

Future Scope

Advanced forecasting

ARIMA

LSTM deep learning

Dashboard & deployment

Power BI or Tableau integration

Streamlit forecasting dashboard

Decision-support visualization layer

Real-time analytics

Live commodity tracking

Automated forecasting pipelines

Continuous model refresh

ROADMAP THEME

Move from offline historical modeling toward real-time, deployable forecasting systems with richer temporal models.

CONCLUSION

From Raw Prices to Forecasting Intelligence

This project shows how machine learning and feature engineering can transform historical commodity data into actionable forecasting insight for analytical and business use.

⏱️ Time-series feature engineering

🔢 Multi-metal regression modeling

📍 Financial trend analysis

🏢 Business-oriented predictive insights

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